

Chapter 1

Matrices

1.1 Matrices and Linear Maps

Matrices are the “coordinated” way to describe linear transformations. We will formalise this in Theorem 2.3 below. But first, we define a linear transformation.

Definition 1.1 (Linear transformation). Fix a field F and two vector spaces V, W over F . A *linear transformation* (or linear map) is a function $T : V \rightarrow W$ such that:

- **Additivity:** $T(v_1 + v_2) = T(v_1) + T(v_2)$ for all $v_1, v_2 \in V$.
- **Homogeneity:** $T(\lambda v) = \lambda T(v)$ for all $\lambda \in F$ and $v \in V$.

Lemma 1.2. *If $T : V \rightarrow W$ is a linear transformation, then $T(0) = 0$.*

Proof. By linearity, $T(0) = T(0 + 0) = T(0) + T(0)$. Adding $-T(0)$ (the additive inverse of $T(0)$ in W) to both sides yields $T(0) = 0$. $\square$

Recall that any finite-dimensional vector space V , with dimension n , is isomorphic to F^n . For now, when we say we have written V and W in coordinates, we mean we replace V and W with F^n and F^m . This motivates the concept of matrices.

Theorem 1.3. *For all linear maps $T : F^n \rightarrow F^m$ defined by $(x_1, \dots, x_n) \mapsto (y_1, \dots, y_m)$, there exists an array of numbers (a_{ij}) such that:*

$$y_i = \sum_{j=1}^n a_{ij} x_j.$$

Proof. We define the standard elementary vectors $e_j \in F^n$ to have value 1 in the j -th entry, and 0 everywhere else. Since $T(e_j)$ is a vector in F^m , we can write it as a list of m coordinates. For all j , define $a_{1j}, \dots, a_{mj} \in F$ by:

$$T(e_j) = \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix}.$$

(Idea: think of $T(e_j)$ as the j^{th} column of the matrix (a_{ij})).

Any input vector can be written as $(x_1, \dots, x_n) = x_1 e_1 + \dots + x_n e_n$ by our definition of addition in F^n . By the linearity of T , we have:

$$\begin{aligned} T(x_1, \dots, x_n) &= T(x_1 e_1 + \dots + x_n e_n) \\ &= x_1 T(e_1) + \dots + x_n T(e_n) \\ &= \sum_{j=1}^n x_j \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix}. \end{aligned}$$

Taking the i^{th} row (coordinate) of the above sum, we get:

$$y_i = \sum_{j=1}^n a_{ij} x_j,$$

as required. □

Remark. The converse is also true: any function $T : F^n \rightarrow F^m$ given by such an array (a_{ij}) is a linear map. This is fundamental: *linear maps are exactly the functions that can be represented by matrices.*

Observe that all the information needed to compute this linear transformation is encoded in the array $A := (a_{ij})$.

Definition 1.4 (Matrix). The set of $m \times n$ matrices over a field F is defined as:

$$M_{m \times n}(F) := \left\{ \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} : a_{ij} \in F \right\}.$$

How do we conveniently encode the action of T using this? We define matrix-vector multiplication:

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$$

The upshot is that, given a linear map $T : F^n \rightarrow F^m$, we *define* matrix multiplication with a column vector precisely to execute this map.

1.2 Matrix Multiplication

1.2.1 Motivation

First, we motivate the definition of general matrix multiplication. Consider two linear maps:

- $T : F^k \rightarrow F^n$ represented by the matrix $B \in M_{n \times k}(F)$
- $S : F^n \rightarrow F^m$ represented by the matrix $A \in M_{m \times n}(F)$

Define the composite linear map $R : F^k \rightarrow F^m$ by $R = S \circ T$. By Claim ($\star$), this new linear map must also have a matrix representation, which we will call $C \in M_{m \times k}(F)$.

We *want* to calculate $c_{i\ell}$, the $(i, \ell)^{\text{th}}$ element of the matrix C . Let's calculate the ℓ^{th} column of C , call it C_ℓ . We know that each column of C represents the effect of the linear map R on the standard basis vector e_ℓ :

$$C_\ell = R(e_\ell) = S(T(e_\ell)).$$

We know $T(e_\ell)$ is exactly the ℓ^{th} column of B , call it B_ℓ . So we must calculate $S(B_\ell)$. Because B_ℓ is a column vector in F^n , evaluating $S(B_\ell)$ is equivalent to taking a linear combination of the columns of A (which we will denote $A_1, \dots, A_n$):

$$S(B_\ell) = b_{1\ell}A_1 + \cdots + b_{n\ell}A_n.$$

To find $c_{i\ell}$, we take the i^{th} row of this result:

$$\begin{aligned} c_{i\ell} &= (C_\ell)_i = b_{1\ell}(A_1)_i + \cdots + b_{n\ell}(A_n)_i \\ &= b_{1\ell}a_{i1} + \cdots + b_{n\ell}a_{in} \\ &= \sum_{j=1}^n a_{ij}b_{j\ell}. \end{aligned}$$

This is exactly the formula for $c_{i\ell}$ that we use to define matrix multiplication. Therefore, the fact that composition of linear maps gives the formula above for the numbers $c_{i\ell}$ is why we define matrix multiplication in the following way:

Definition 1.5 (Matrix operations).

- Matrix multiplication is the function $\cdot : M_{m \times n}(F) \times M_{n \times p}(F) \rightarrow M_{m \times p}(F)$ defined by:

$$(a_{ij})_{m \times n} (b_{kl})_{n \times p} := \left(c_{il} = \sum_{j=1}^n a_{ij} b_{jl} \right)_{m \times p}$$

- Matrix addition is the function $+ : M_{m \times n}(F) \times M_{m \times n}(F) \rightarrow M_{m \times n}(F)$ defined by:

$$(a_{ij}) + (b_{ij}) = (a_{ij} + b_{ij}).$$

- Scalar multiplication for matrices is the function $\cdot : F \times M_{m \times n}(F) \rightarrow M_{m \times n}(F)$ defined by:

$$\lambda(a_{ij}) = (\lambda a_{ij}).$$

Remark. Just like $F^n \cong M_{n \times 1}(F)$, the set $M_{m \times n}(F)$ forms a vector space under matrix addition and scalar multiplication. However, matrix multiplication gives $M_{n \times n}(F)$ *additional* structure beyond just a vector space (it forms an algebra).

Definition 1.6 (Square Matrix). A matrix is square if $A \in M_{n \times n}(F)$ (the number of rows equals the number of columns). Square matrices of a fixed size can be multiplied by each other.

Proposition 1.7. *Matrix multiplication is associative: $(AB)C = A(BC)$.*

Proof.

$$((AB)C)_{il} = \sum_j (AB)_{ij} c_{jl} = \sum_j \left(\sum_k a_{ik} b_{kj} \right) c_{jl} = \sum_{j,k} a_{ik} b_{kj} c_{jl}$$

$$(A(BC))_{il} = \sum_k a_{ik} (BC)_{kl} = \sum_k a_{ik} \left(\sum_j b_{kj} c_{jl} \right) = \sum_{j,k} a_{ik} b_{kj} c_{jl}$$

Therefore, $((AB)C)_{il} = (A(BC))_{il}$. □

Corollary 1.8. *Given linear maps $S : F^m \rightarrow F^n$ and $T : F^n \rightarrow F^p$ with matrix representations A and B (so $S(v) = Av$ and $T(v) = Bv$), the matrix representation of the composite map $T \circ S : F^m \rightarrow F^p$ is given by the matrix product BA .*

Proof. $T \circ S(v) = T(S(v)) = B(Av) = (BA)v$ by associativity. $\square$

1.2.2 Some more formal properties of matrices

Definition 1.9 (Bilinear map). Let V, W, U be vector spaces over F . A map $f : V \times W \rightarrow U$ mapping $(v, w) \mapsto f(v, w)$ is *bilinear* if it is linear in each argument when the other is held fixed:

- $(\lambda_1 v_1 + \lambda_2 v_2, w) \mapsto \lambda_1 f(v_1, w) + \lambda_2 f(v_2, w)$
- $(v, \lambda_1 w_1 + \lambda_2 w_2) \mapsto \lambda_1 f(v, w_1) + \lambda_2 f(v, w_2)$

Remark. Note that this implies the standard scalar extraction property $f(\lambda v, w) = f(v, \lambda w) = \lambda f(v, w)$ (simply set $\lambda_2 = 0$).

Example 1.10. The dot product $F^n \times F^n \rightarrow F$, defined by $((a_1, \dots, a_n), (b_1, \dots, b_n)) \mapsto a_1 b_1 + \dots + a_n b_n$ is bilinear. This follows easily from the distributive properties of the field F .

We want to prove that matrix multiplication is bilinear. To do this, we first establish how matrix multiplication interacts with columns and rows.

Proposition 1.11. *Let $A \in M_{m \times n}(F)$ and $B = (B_1, \dots, B_p) \in M_{n \times p}(F)$, where B_j are the columns of B . Then:*

$$AB = (AB_1, \dots, AB_p)$$

Proof. By the formula, $(AB)_{i\ell} = \sum_k a_{ik} b_{k\ell}$. Notice that $\sum_k a_{ik} b_{k\ell}$ is exactly the i^{th} element of the matrix-vector product AB_ℓ . Thus, the ℓ^{th} column of AB is exactly the vector AB_ℓ . $\square$

Remark. This provides a *column-wise* way to calculate matrix products. To find the entire j^{th} column of AB , just calculate the matrix-vector product AB_j . The j^{th} column of AB **only** depends on the j^{th} column of B .

Proposition 1.12. *We can separate rows under right multiplication. Let B_i be the rows of B . Then:*

$$\begin{pmatrix} B_1 \\ \vdots \\ B_p \end{pmatrix} A = \begin{pmatrix} B_1 A \\ \vdots \\ B_p A \end{pmatrix}$$

Proof. Same logic as above, switching indices appropriately. $\square$

Corollary 1.13. *Matrix multiplication is bilinear.*

Proof. We prove linearity in the second argument. Let A be an $m \times n$ matrix, B and C be $n \times p$ matrices, and $\lambda \in F$.

$$\begin{aligned} A(\lambda B) &= A(\lambda B_1 \dots \lambda B_n) \\ &= (A(\lambda B_1) \dots A(\lambda B_n)) && \text{(Column separation)} \\ &= (\lambda(AB_1) \dots \lambda(AB_n)) && \text{(Matrix-vector linearity)} \\ &= \lambda(AB_1 \dots AB_n) \\ &= \lambda(AB) \end{aligned}$$

And for addition:

$$\begin{aligned} A(B + C) &= A(B_1 + C_1 \dots B_n + C_n) \\ &= (A(B_1 + C_1) \dots A(B_n + C_n)) \\ &= (A(B_1) + A(C_1) \dots A(B_n) + A(C_n)) \\ &= (A(B_1) \dots A(B_n)) + (A(C_1) \dots A(C_n)) \\ &= AB + AC \end{aligned}$$

Linearity in the first argument follows similarly. $\square$

Definition 1.14 (Transpose). The transpose of a matrix $A = (a_{ij}) \in M_{m \times n}(F)$ is defined as $A^\top := (a_{ji}) \in M_{n \times m}(F)$.

Proposition 1.15. *The transpose operation satisfies the following properties:*

- *Transpose is linear:* $(A + B)^\top = A^\top + B^\top$ and $(\lambda A)^\top = \lambda A^\top$.

- $(A^\top)^\top = A$.
- $(AB)^\top = B^\top A^\top$.

Proof. We prove the product rule. Let $C = AB$. Then $c_{ij} = \sum_k a_{ik}b_{kj}$. The (i, j) entry of $(AB)^\top$ is the (j, i) entry of AB , which is $\sum_k a_{jk}b_{ki}$. The (i, j) entry of $B^\top A^\top$ is the dot product of the i -th row of $B^\top$ and the j -th column of $A^\top$. This is exactly $\sum_k (B^\top)_{ik}(A^\top)_{kj} = \sum_k b_{ki}a_{jk} = \sum_k a_{jk}b_{ki}$. $\square$

Remark. Notice that the order reverses in $(AB)^\top = B^\top A^\top$. This is necessary for the dimensions to align. For example, if A is 1×5 and B is 5×1 , AB is 1×1 . $A^\top$ is 5×1 and $B^\top$ is 1×5 . The product $A^\top B^\top$ is impossible, but $B^\top A^\top$ results in a 1×1 matrix. There is a deeper reason for this, which we will see when we look at dual vector spaces and dual linear transformations.

1.2.3 Special Matrices

Definition 1.16 (Symmetric Matrix). A square matrix A is *symmetric* if $A = A^\top$.

Definition 1.17 (Orthogonal Matrix). A square matrix A is *orthogonal* if $AA^\top = I$.

Remark. For a square matrix, $AA^\top = I \iff A^\top A = I$. What does it mean for $AA^\top = I$? We will discuss this in more detail in a later chapter, but for now it is enough to note that if A_i are the rows of A , then $(AA^\top)_{ij} = A_i \cdot A_j$. Therefore:

$$A_i \cdot A_j = \delta_{ij} := \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

This means the *rows* of A are orthonormal. Similarly, $A^\top A = I$ implies the *columns* of A are orthonormal. Geometrically in $\mathbb{R}^n$, multiplying by an orthogonal matrix represents a rigid transformation (a rotation or reflection) that preserves lengths and angles (again, this will make more sense in a later chapter).

Definition 1.18 (Triangular and Diagonal Matrices). Let $A = (a_{ij})$ be a matrix.

- A is *upper-triangular* if $a_{ij} = 0$ for all $i > j$.

- A is *lower-triangular* if $a_{ij} = 0$ for all $i < j$.
- A is *diagonal* if it is both upper and lower triangular (i.e., $a_{ij} = 0$ for all $i \neq j$).